RTL Final Report

February 7, 2026

Project Summary

This project presents the design and implementation of a robust, user-configurable multi-clock generation and switching system with strong emphasis on data integrity, operational safety, and runtime controllability. The system is implemented as an AXI4-Lite–controlled hardware block capable of generating multiple clock frequencies, safely switching between them, and reliably transferring data across asynchronous clock domains using well-defined synchronization mechanisms. The design specifically targets FPGA-based embedded systems, where dynamic frequency selection and dependable clock-domain crossing are critical requirements.

The primary objective of the design is to allow an external processor to dynamically select one of seven generated clock frequencies while ensuring that no data corruption occurs during clock transitions. This is achieved through the coordinated operation of asynchronous FIFO buffers, a dedicated clock-switching finite state machine (FSM), and a programmable control plane exposed through an AXI4-Lite interface. The overall architecture emphasizes modularity, safety, and observability, making it well-suited for both functional verification and future scalability.

System Architecture Overview

The complete system is organized as an integrated clock-control and data-handling subsystem that operates as an AXI4-Lite slave. The architecture is divided into four major functional blocks: clock generation, data path, control plane, and software interface.

The clock generation unit produces seven configurable lower-frequency clocks derived from a single high-speed system clock using parameterized divider logic. These generated clocks serve as independent write-side clocks for the data path. To ensure low skew and stable clock distribution, each generated clock is routed through global clock buffers.

The data path consists of seven asynchronous FIFO buffers, each associated with a distinct generated write clock while sharing a common read clock driven by the stable system clock. This architecture enables safe and deterministic transfer of data from multiple asynchronous clock domains into a single synchronous domain. FIFO-based clock-domain crossing mitigates metastability risks by synchronizing read and write pointers using established techniques such as multi-flop synchronization and Gray-coded pointers.

The control plane is implemented using a dedicated finite state machine responsible for safe clock switching. Upon receiving a frequency-change request from software, the FSM temporarily halts FIFO read and write operations and waits until all FIFOs are fully drained. Only after this condition is satisfied does the FSM enable the clock transition. Following the switch, a stabilization period is enforced before resuming normal operation to ensure that the new clock domain is fully

settled. This controlled sequence prevents data loss, metastability, and undefined timing behavior during runtime frequency changes.

The AXI4-Lite interface provides the software-visible configuration and monitoring layer. Through memory-mapped registers, an external processor can select the active FIFO, initiate clock-switch requests, read FIFO data, and monitor system status. In addition, the interface exposes diagnostic and telemetry registers that enable real-time verification of clock frequencies and internal system states.

Operational Mechanisms and Safety Features

The asynchronous FIFO mechanism forms the core of the system, enabling reliable communication between independent clock domains. Data is written into the FIFO using a variable-frequency write clock and read out using a fixed system clock. FIFO control logic ensures that write operations occur only when the buffer is not full and read operations occur only when data is available, thereby preventing overflow and underflow conditions.

The multi-clock FIFO subsystem extends this concept by instantiating seven independent FIFOs, each driven by a different generated clock. A selection signal determines which FIFO output is forwarded to downstream logic. Integrated clock-edge counters are included for each generated clock, allowing runtime verification of frequency behavior through software-readable registers.

Clock switching is governed by a safety-critical FSM that enforces strict sequencing rules. Clock changes are permitted only when all FIFOs are empty, ensuring that no in-flight data is lost. During the transition, FIFO operations are temporarily disabled, and normal operation is resumed only after the new clock has stabilized. This design guarantees deterministic behavior even under frequent or rapid frequency-switch requests.

To further enhance system robustness, reset interlocking logic is employed to ensure that FIFO state remains stable during clock transitions. All critical control and status signals are aggregated and reported through the AXI-Lite interface, providing comprehensive visibility into system operation for debugging and validation.

Findings, Design Improvements, and Significance

The final implementation successfully demonstrates a safe and flexible runtime clock-switching architecture capable of preserving data integrity across asynchronous clock domains. By combining asynchronous FIFOs, structured FSM-based control, and a standardized AXI4-Lite interface, the system achieves a high degree of configurability, reliability, and observability. The inclusion of diagnostic counters and status registers supports both hardware-level and software-driven verification of correct system behavior.

Compared to simpler clock-selection approaches, this design significantly improves reliability by explicitly addressing clock-domain crossing hazards and unsafe frequency transitions. The modular architecture also facilitates future enhancements, such as replacing counter-based clock division with PLL/MMCM-based clock generation, enabling dynamic frequency configuration through software, and incorporating advanced fault detection and error-handling mechanisms.

Overall, this project establishes a strong foundation for a production-quality clock management and multi-frequency data processing subsystem suitable for FPGA-based real-time and embedded applications. The design demonstrates sound engineering practices in clock-domain crossing, runtime control, and safety-critical hardware design, while remaining scalable and extensible for future development.

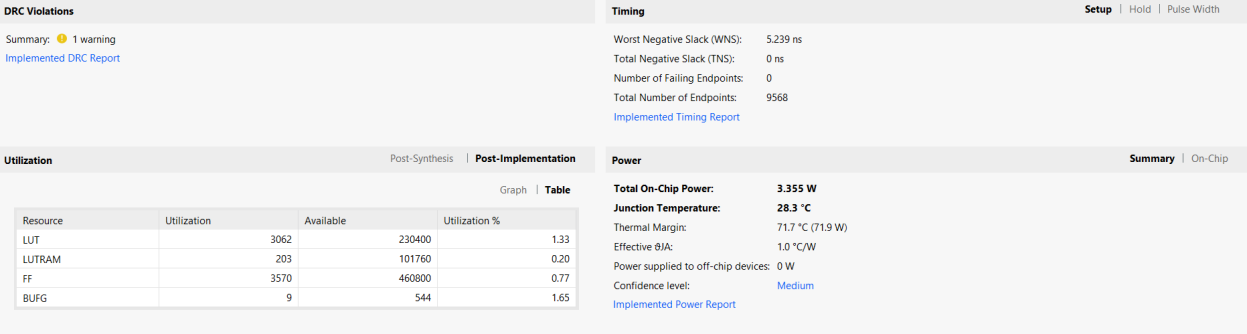


Figure 1: Initial post-implementation results highlighting higher resource utilization, multiple global clock buffers, and timing complexity associated with logic-generated clocks.

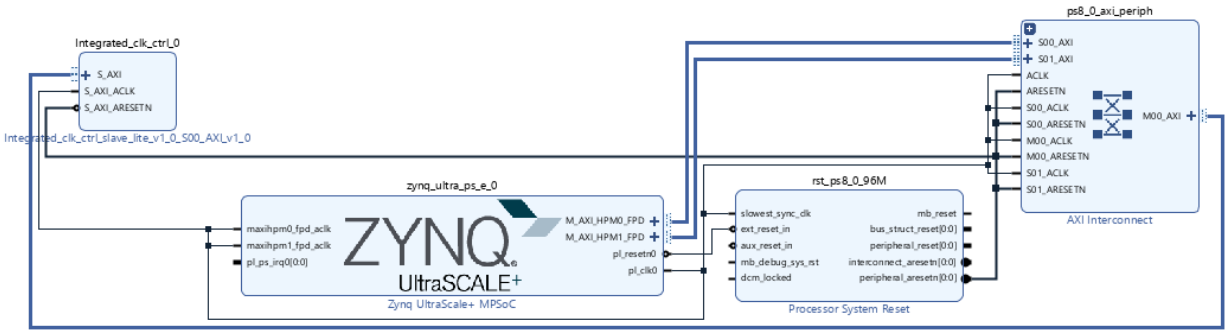


Figure 2: Top-level system integration of the AXI4-Lite–controlled multi-clock FIFO subsystem within the Zynq UltraScale+ processing system.

Design Evolution and Implemented Improvements

This section describes the progressive enhancements made to the clock control and multi-channel FIFO system to improve synthesis compatibility, robustness, power efficiency, observability, and timing safety. The improvements were carried out iteratively, addressing architectural limitations, verification challenges, and FPGA implementation constraints, ultimately resulting in a production-ready and scalable design.

Architectural Refactoring and FPGA Compliance

The initial implementation relied on multiple logic-generated clocks, gated reset logic, and asynchronous FIFO structures to support dynamic frequency control. Although functionally correct, this approach introduced synthesis violations, excessive clock-domain crossing (CDC) complexity, and difficulties in achieving timing closure on FPGA platforms.

To resolve these issues, the architecture was refactored to adopt a single-system-clock, clock-enable–based design methodology. All internally generated clocks were replaced with per-channel clock-enable pulses derived from programmable counters, while a single global system clock was retained. In addition, dynamic reset gating was removed and replaced with a clean, static global reset signal.

The FIFO architecture was also redesigned from a multi-clock asynchronous structure into a

synchronous, clock-enable–controlled FIFO system. All FIFO operations now occur synchronously with the system clock, eliminating CDC hazards and simplifying both verification and timing analysis.

Outcome:

- Full compliance with FPGA synthesis guidelines
- Elimination of clock-domain crossing risks
- Simplified timing closure
- Improved design clarity and maintainability

Debugging, Integration Corrections, and FIFO Refinement

Following architectural refactoring, several integration and synthesis issues were identified during system-level compilation and simulation. These included port name mismatches, signal width inconsistencies, incorrect FIFO interface assumptions, and incomplete AXI4-Lite protocol handling in the testbench.

The FIFO subsystem was further refined into a single-clock, multi-channel FIFO architecture with independent read and write pointers for each channel, controlled via clock-enable signals. All port mismatches were corrected, and FIFO status signals were consistently handled as scalar outputs corresponding to the selected channel.

The finite state machine (FSM) responsible for frequency control was enhanced by introducing an explicit error flag and expanding the FSM status bus. These changes improved system observability and simplified runtime debugging. AXI4-Lite interface logic was also corrected to ensure full protocol compliance, including proper response signaling.

Outcome:

- Error-free synthesis and simulation
- Deterministic FIFO behavior
- Enhanced FSM visibility
- Reliable AXI-based configuration and control

Verification Strategy Alignment

With the transition to a clock-enable–driven architecture, traditional verification approaches that relied on probing internally generated clocks were no longer valid. The verification strategy was therefore updated to align with the final hardware implementation.

Channel activity and effective frequency behavior were verified using clock-edge counters rather than internal clock signals. Frequency changes were validated by observing counter increments over fixed observation windows, ensuring correctness while avoiding invalid hierarchical signal access.

Outcome:

- Architecture-consistent verification methodology
- Tool-safe and scalable testing approach
- Increased confidence in functional correctness

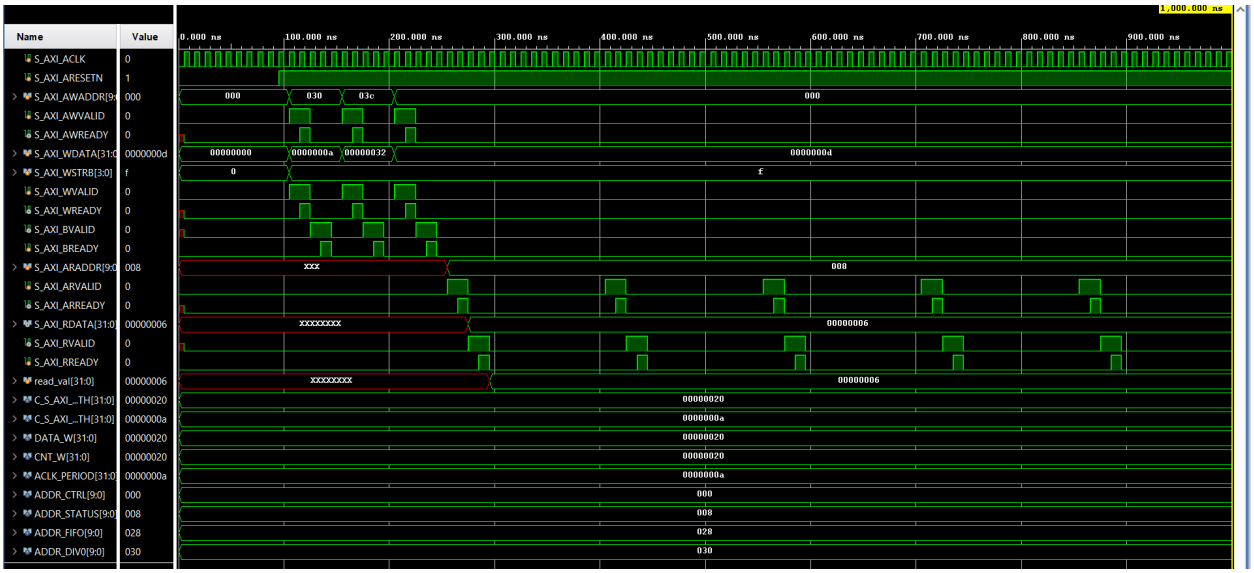


Figure 3: Functional simulation waveforms demonstrating correct AXI4-Lite transactions, FIFO data transfers, and controlled clock-switch sequencing under the initial architecture.

Modular Clock Gating and Power Optimization

To improve power efficiency and modularity, a dedicated clock gating unit was introduced. Instead of gating clock signals directly, an FPGA-safe clock-enable–based gating mechanism was adopted.

Raw clock-enable pulses generated for each channel are gated using a combination of software-programmable per-channel gate masks and a global enable signal driven by the FSM. This ensures that clock activity occurs only when permitted by both software and system-level safety conditions.

The clock gating logic was centralized into a standalone module and removed from the FIFO subsystem, resulting in a clear separation of concerns. The AXI-Lite interface was extended to allow runtime software control over individual channel activation.

Outcome:

- Reduced dynamic power consumption
- Improved modularity and readability
- Safe interaction between software control and FSM authority

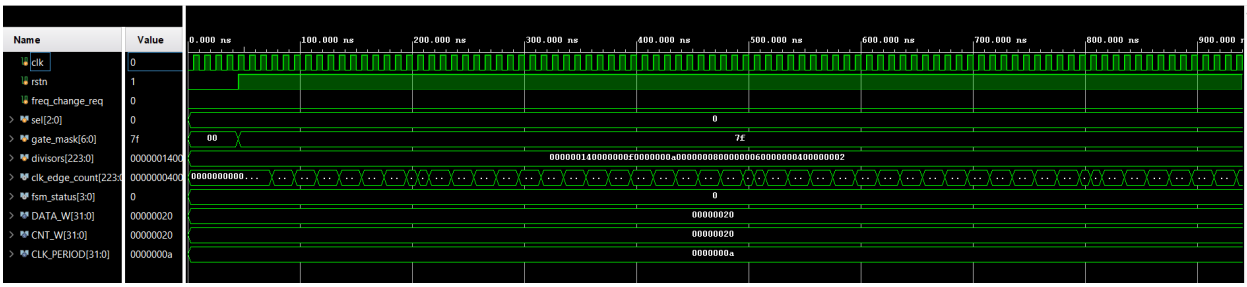


Figure 4: Simulation results illustrating FSM-controlled clock-enable gating, safe channel activation, and deterministic FIFO operation during runtime frequency changes.

Enhanced Observability Through Sticky Telemetry

In earlier versions of the design, FIFO overflow and underflow conditions were indicated using transient pulse signals, which could be missed by software polling. To improve observability, persistent sticky telemetry registers were introduced.

Sticky registers capture FIFO overflow and underflow events on a per-channel basis and retain this information until explicitly cleared by software. These registers were mapped to read-only AXI-Lite status registers alongside live FIFO status signals.

Outcome:

- Persistent fault logging
- Improved post-event debugging capability
- Clear separation between real-time status and historical error data

Timing Constraints and Multi-Clock Safety

Although the final design uses a single system clock for logic operation, internally derived timing signals and AXI clock domains must be explicitly defined for correct synthesis and implementation. A timing constraints file was therefore introduced to define all relevant clock domains.

Asynchronous clock group constraints were applied to prevent false timing analysis between unrelated domains, and false path constraints were added for asynchronous reset signals. These constraints ensure accurate timing analysis and reliable timing closure.

Outcome:

- Elimination of false timing violations
- Improved synthesis and implementation reliability
- Safe and deterministic multi-clock operation

Final Outcome

Through systematic architectural refinement, debugging, verification alignment, power optimization, observability enhancement, and timing constraint definition, the design evolved into a robust, FPGA-compliant, and production-ready clock control system. The final implementation demonstrates strong engineering practices in clock management, AXI-based control, fault monitoring, and timing safety, while remaining scalable and extensible for future enhancements.

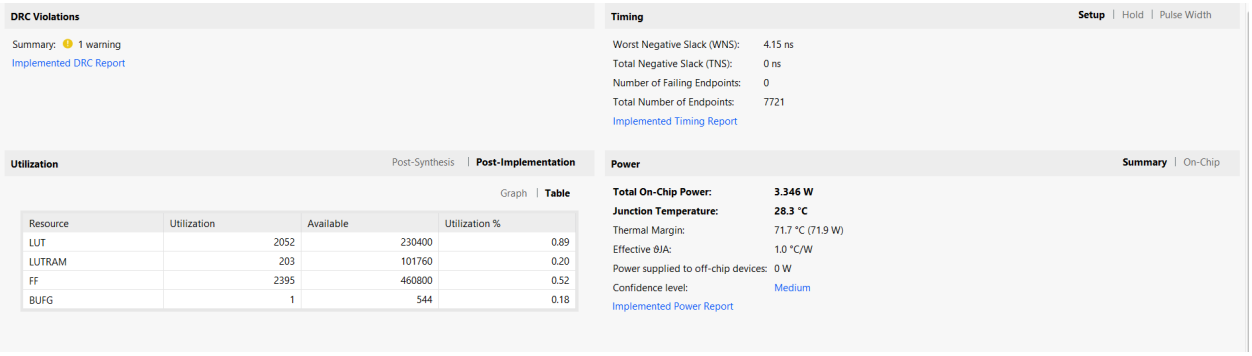


Figure 5: Final post-implementation timing, power, and utilization results showing zero timing violations, reduced resource usage, and improved clocking efficiency in the refactored design.

Table 1: Post-Implementation Results Comparison

Metric	**Initial Design**	**Final Design**
LUT Utilization	3062 (1.33%)	2052 (0.89%)
Flip-Flop (FF) Utilization	3570 (0.77%)	2395 (0.52%)
LUTRAM Utilization	203 (0.20%)	203 (0.20%)
BUFG Usage	9 (1.65%)	1 (0.18%)
Worst Negative Slack (WNS)	5.239 ns	4.15 ns
Total Negative Slack (TNS)	0 ns	0 ns
Failing Endpoints	0	0
Total Endpoints	9568	7721
Total On-Chip Power	3.355 W	3.346 W
Junction Temperature	28.3°C	28.3°C